EXPERIMENT-4

Write a program of Missionaries and Cannibals using python code

Aim : To solve the Missionaries and Cannibals problem using Python.

Procedure:

Represent the number of missionaries and cannibals on the left bank.

Move people across the river using a boat of capacity 2.

Ensure cannibals never outnumber missionaries on either bank.

Use Breadth-First Search (BFS) to find a valid sequence of moves.

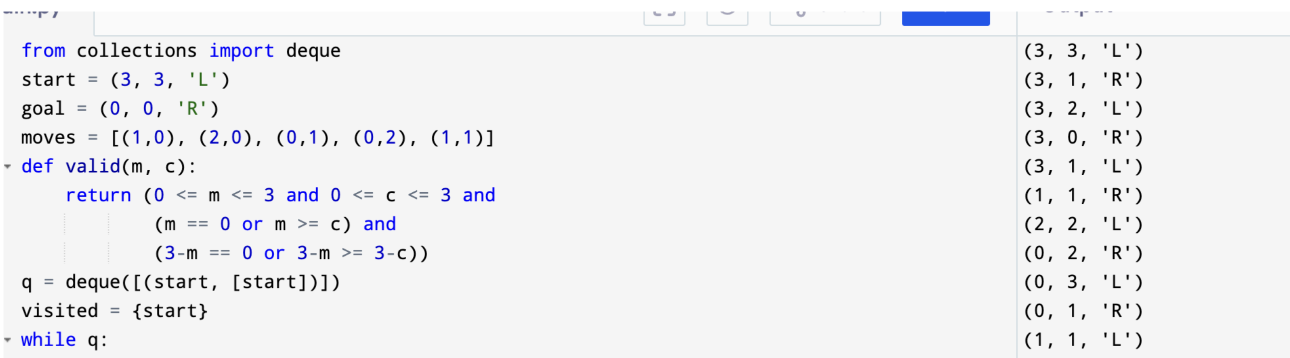
Print the steps from the start state to the goal state.

Python Code:

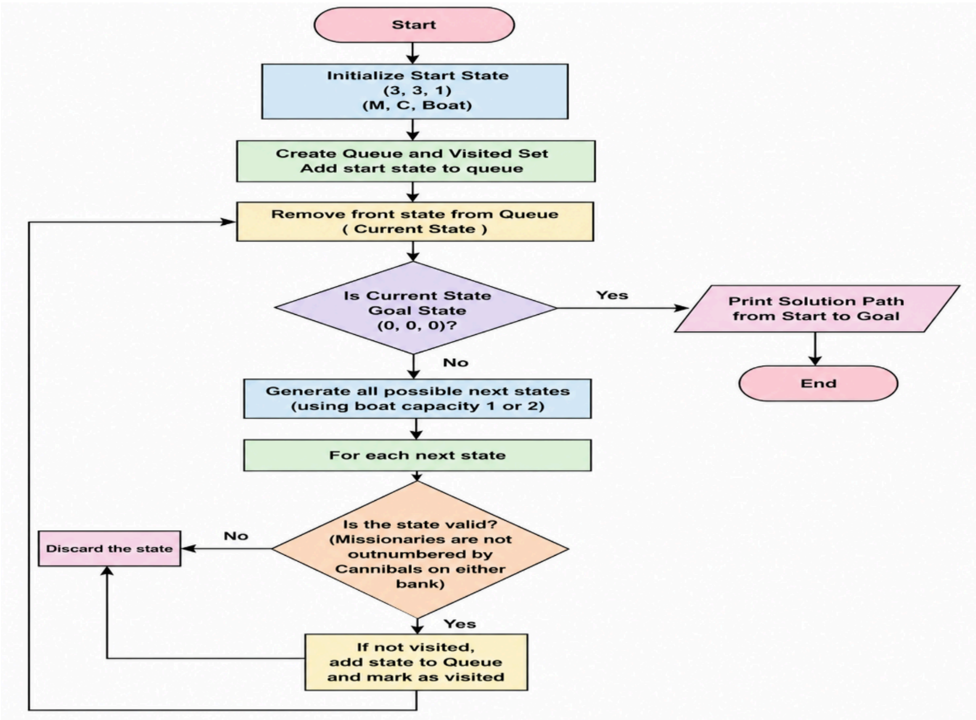
```
from collections import deque

start = (3, 3, 1)      # (Missionaries, Cannibals, Boat)

goal = (0, 0, 0)

moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]

def valid(state):

    m, c, b = state

    if m < 0 or c < 0 or m > 3 or c > 3:

        return False

    if (m > 0 and m < c):

        return False

    if ((3-m) > 0 and (3-m) < (3-c)):

        return False

    return True
```

```
queue = deque([(start, [start])])
visited = set()
while queue:
    state, path = queue.popleft()
    if state == goal:
        print("Solution Path:")
        for step in path:
            print(step)
        break
    if state in visited:
        continue
    visited.add(state)
    m, c, b = state
    for dm, dc in moves:
        if b == 1:
            new = (m-dm, c-dc, 0)
        else:
            new = (m+dm, c+dc, 1)
        if valid(new):
            queue.append((new, path+[new]))
```

Output:



Flowchart:



Result: Hence, Missionaries and Cannibals problem has solved